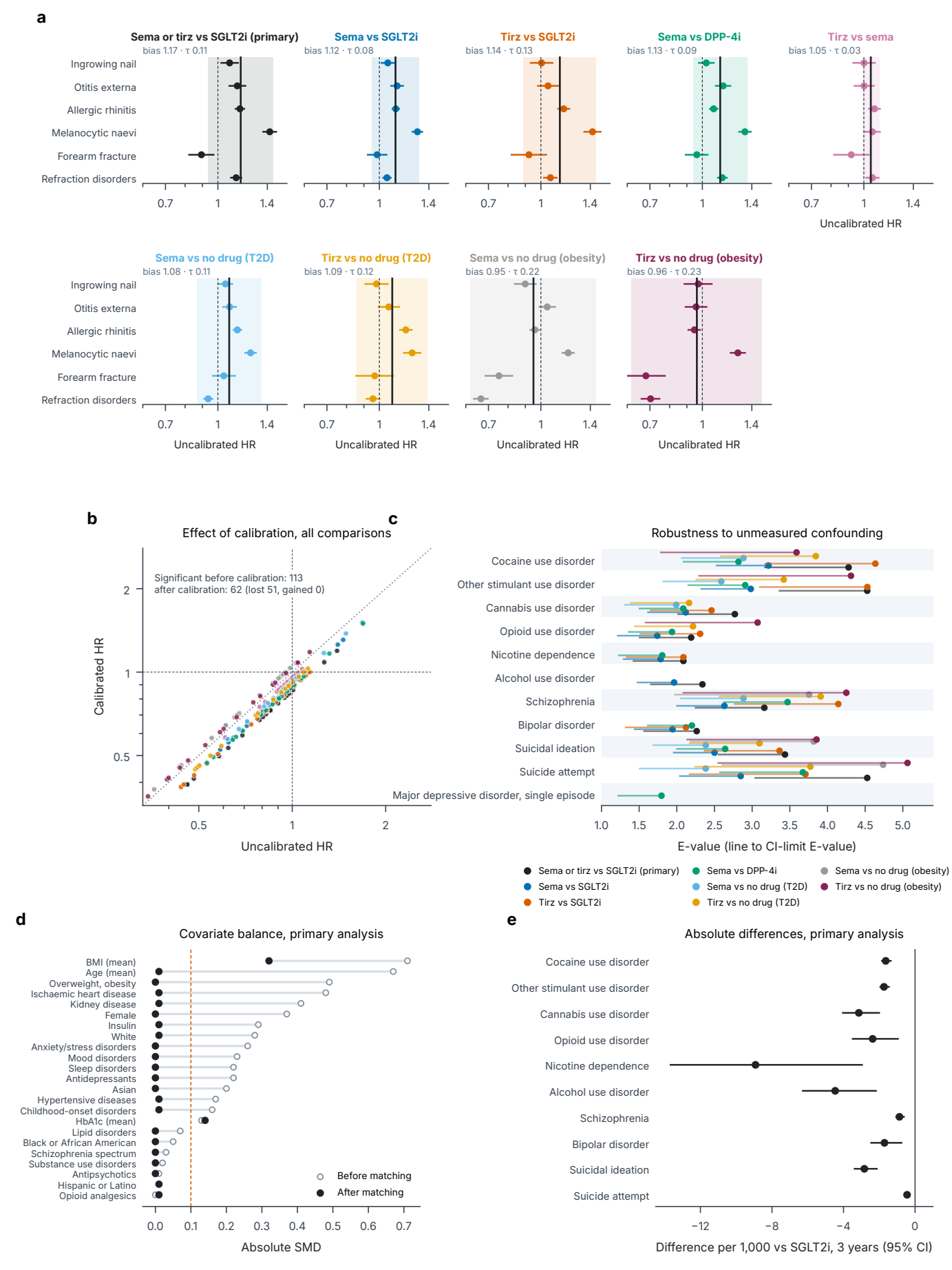

a
Sema or tirz vs SGLT2i (primary)
bias 1.17 · τ 0.11
Sema vs SGLT2i
bias 1.12 · τ 0.08
Tirz vs SGLT2i
bias 1.14 · τ 0.13
Sema vs DPP-4i
bias 1.13 · τ 0.09
Tirz vs sema
bias 1.05 · τ 0.03
Ingrowing nail
Otitis externa
Allergic rhinitis
Melanocytic naevi
Forearm fracture
Refraction disorders
0.7
1
1.4
Uncalibrated HR
Sema vs no drug (T2D)
bias 1.08 · τ 0.11
Tirz vs no drug (T2D)
bias 1.09 · τ 0.12
Sema vs no drug (obesity)
bias 0.95 · τ 0.22
Tirz vs no drug (obesity)
bias 0.96 · τ 0.23
b
Effect of calibration, all comparisons
Significant before calibration: 113
after calibration: 62 (lost 51, gained 0)
Calibrated HR
Uncalibrated HR
0.5
1
2
c
Robustness to unmeasured confounding
Cocaine use disorder
Other stimulant use disorder
Cannabis use disorder
Opioid use disorder
Nicotine dependence
Alcohol use disorder
Schizophrenia
Bipolar disorder
Suicidal ideation
Suicide attempt
Major depressive disorder, single episode
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
E-value (line to CI-limit E-value)
Sema or tirz vs SGLT2i (primary)
Sema vs SGLT2i
Tirz vs SGLT2i
Sema vs DPP-4i
Sema vs no drug (T2D)
Tirz vs no drug (T2D)
Sema vs no drug (obesity)
Tirz vs no drug (obesity)
d
Covariate balance, primary analysis
BMI (mean)
Age (mean)
Overweight, obesity
Ischaemic heart disease
Kidney disease
Female
Insulin
White
Anxiety/stress disorders
Mood disorders
Sleep disorders
Antidepressants
Asian
Hypertensive diseases
Childhood-onset disorders
HbA1c (mean)
Lipid disorders
Black or African American
Schizophrenia spectrum
Substance use disorders
Antipsychotics
Hispanic or Latino
Opioid analgesics
Before matching
After matching
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Absolute SMD
e
Absolute differences, primary analysis
Cocaine use disorder
Other stimulant use disorder
Cannabis use disorder
Opioid use disorder
Nicotine dependence
Alcohol use disorder
Schizophrenia
Bipolar disorder
Suicidal ideation
Suicide attempt
−12
−8
−4
0
Difference per 1,000 vs SGLT2i, 3 years (95% CI)